

Physics and Astronomy REU: Texas A&M University-Commerce

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University: TAMUC
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TEXAS A&M UNIVERSITY
COMMERCE



Outline

- Background
- Goal
- Target Selection
- Observations
- Methods
- Analysis
- Results



Image Credit: TAMUC

Stellar Evolution

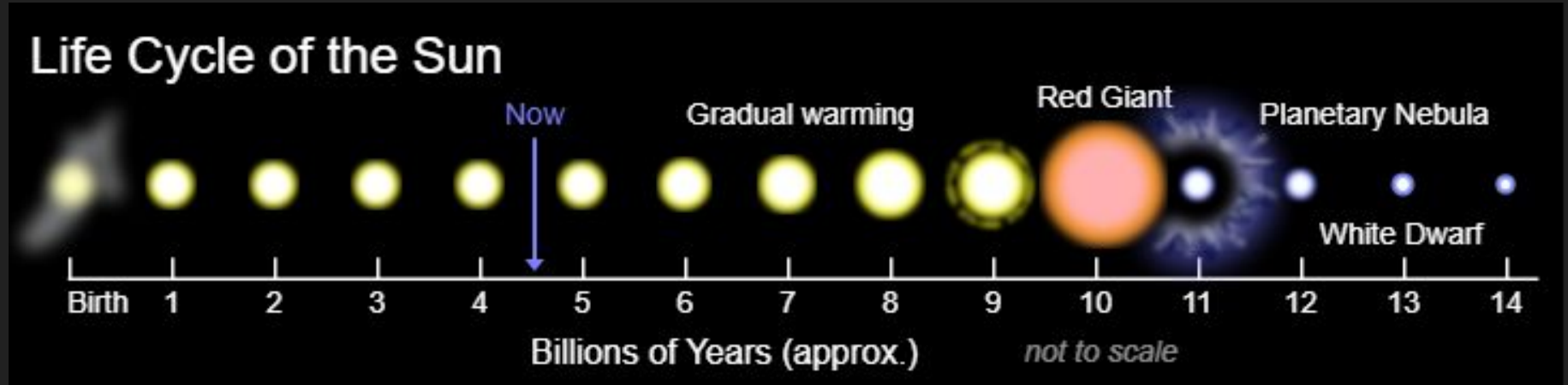


Image Credit: David Jeffery; www.physics.unlv.edu

White Dwarf Stars (WDs)

- $M \leq 1.4 M_{\odot}$
- $R \approx R_{\oplus}$
- Larger Mass = Smaller Radius
- $M_{SB} = 1.02 M_{\odot}$
- $M_{PB} = 0.6 M_{\odot}$
- $M_{VMS} = 0.7 M_{\odot}$

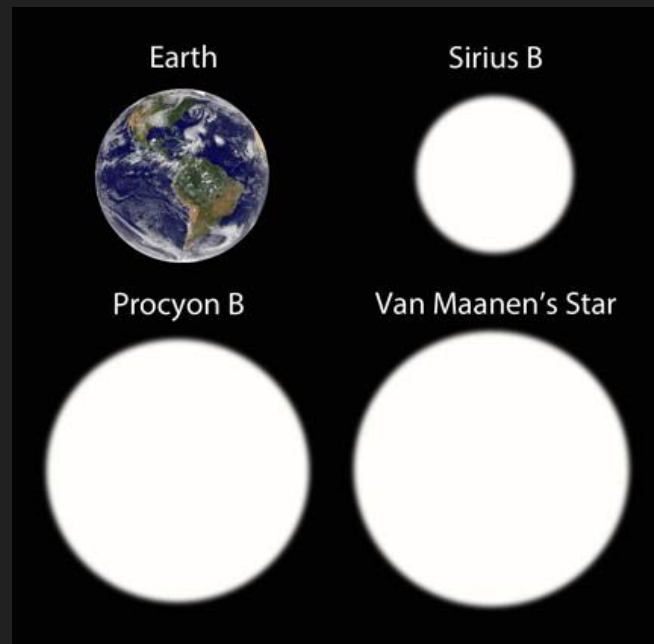


Image Credit: Bob King; skyandtelescope.org

My Project

Goal: Search for variation and rapid rotation in massive WDs.

- Why might it vary in luminosity?
 - Starspots
- Why might it rapidly rotate?
 - Massive Sun-like star
 - Double Degenerate Merger

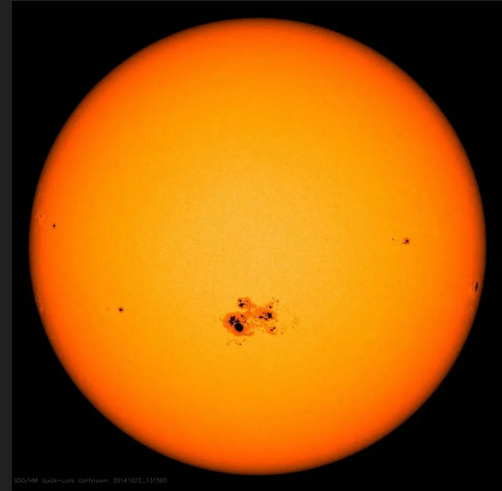
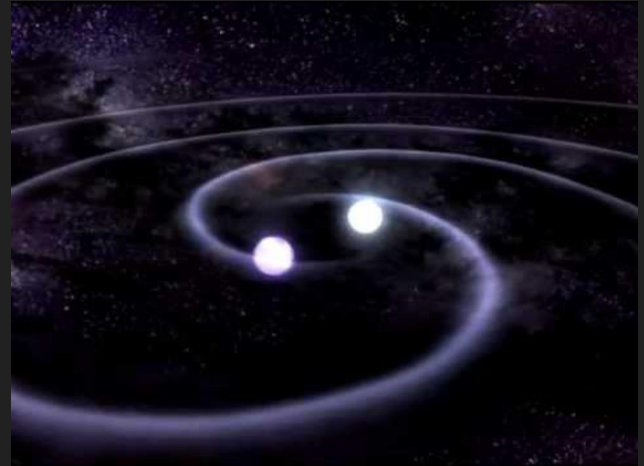


Image credit: NASA/SDO

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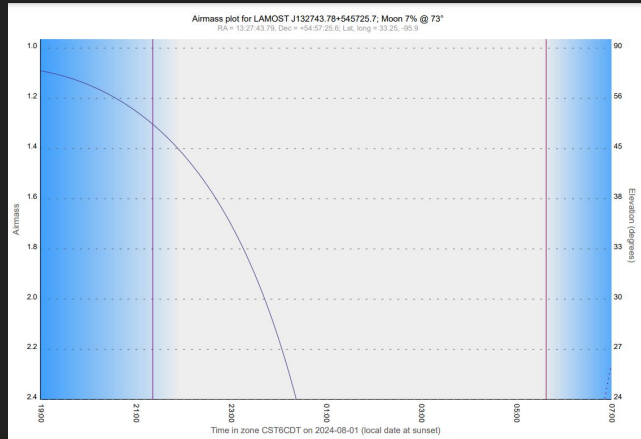


Video credit: astropage.eu

Credit: NASA/Tod Strohmayer (GSFC)/Dana Berry (Chandra X-Ray Observatory)

Target Selection

- Montreal White Dwarf Database
- MAST TESS Archive
- Parameters for MWDD
- Airmass Plot



Montreal White Dwarf Database

Home Tables and charts References Database description Evolutionary tracks

Search filters

	Minimum	Maximum	Spectral Type
T_{eff} [K]:		T_{eff} [K]:	☒ DA 12
$\log(g)$:		$\log(g)$:	☒ DB 0
distance [pc]:		distance [pc]:	☒ DC 0
Mass v	0.9	G v 16.5	☒ DQ 1
			☒ DZ 0
			☒ DO 0
			☐ PG1159 0
			☒ CND? 0
			☐ Subdwarf 0
			☐ Other 0

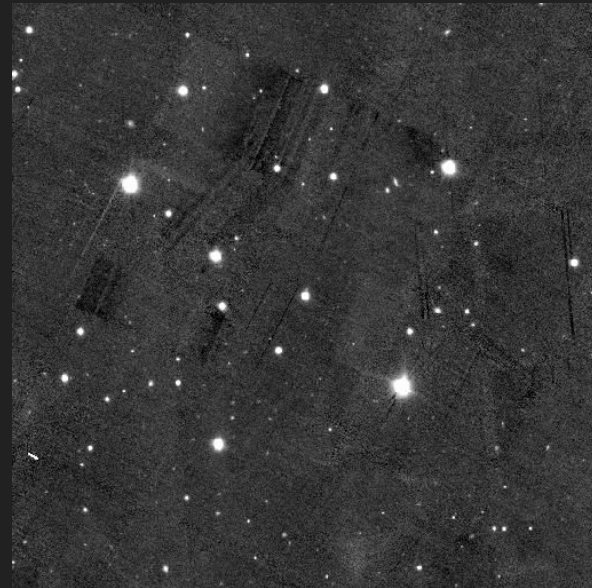
Total: 13
(13 including unresolved double degenerates)

Radial search

Right ascension	Declination	Distance
hh mm ss.ss	dd mm ss.ss	arcmin

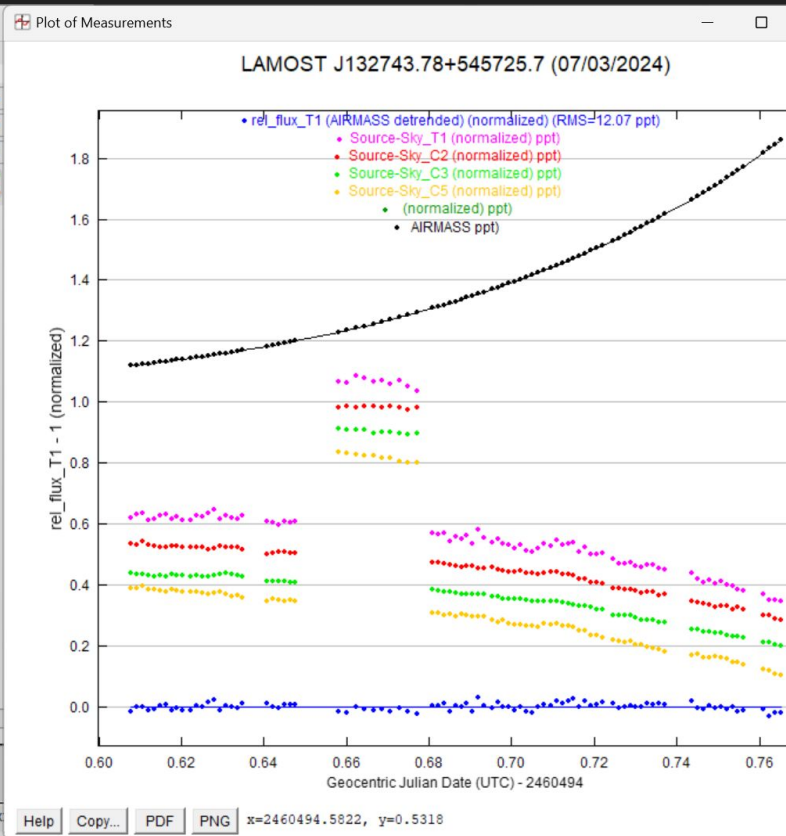
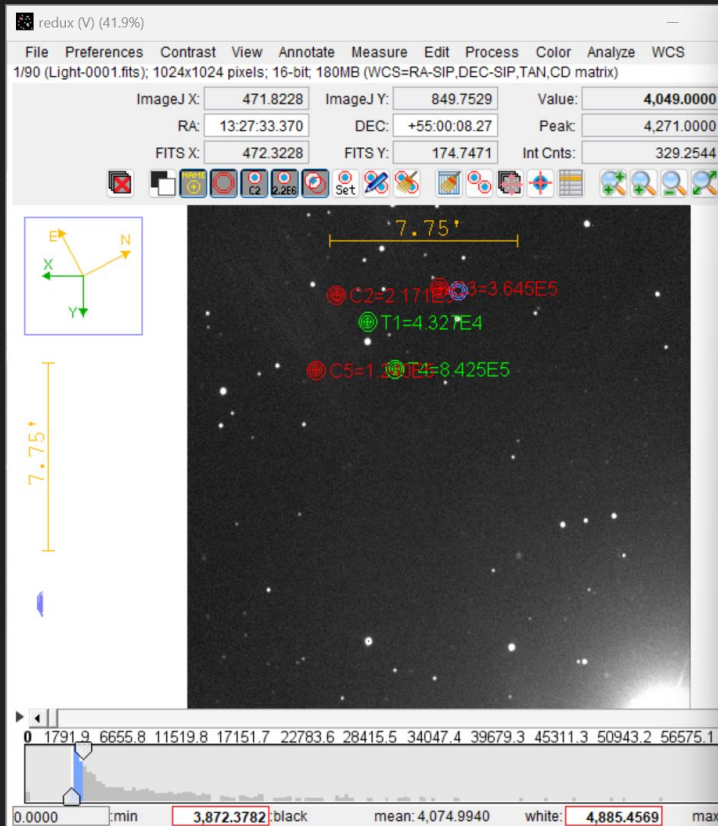
Stars Observed

- PG 1533-057
 - Obs 2 days
 - RA: 15 36 34.92 ,DEC: -05 52 39.73
 - $M \approx 0.94 M_{\odot}$
- LAMOST J132743.78+545725.7
 - Obs 4 days
 - RA: 13 27 43.79 ,DEC: 54 57 25.67
 - $M \approx 1.05 M_{\odot}$
- WD 1459+347
 - Obs 2 days
 - RA: 15 01 41.07 ,DEC: 34 31 59.34
 - $M \approx 0.925 M_{\odot}$



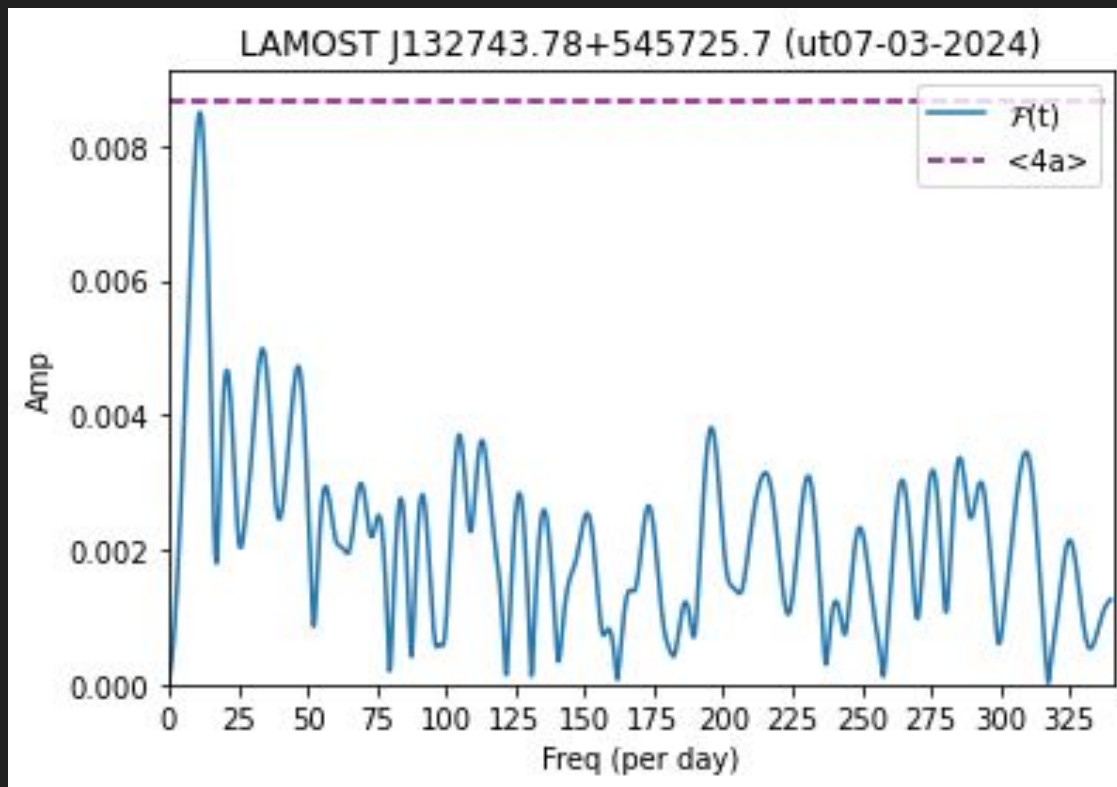
Info: MWDD, Image Credit: PanSTARRS-1

Photometry

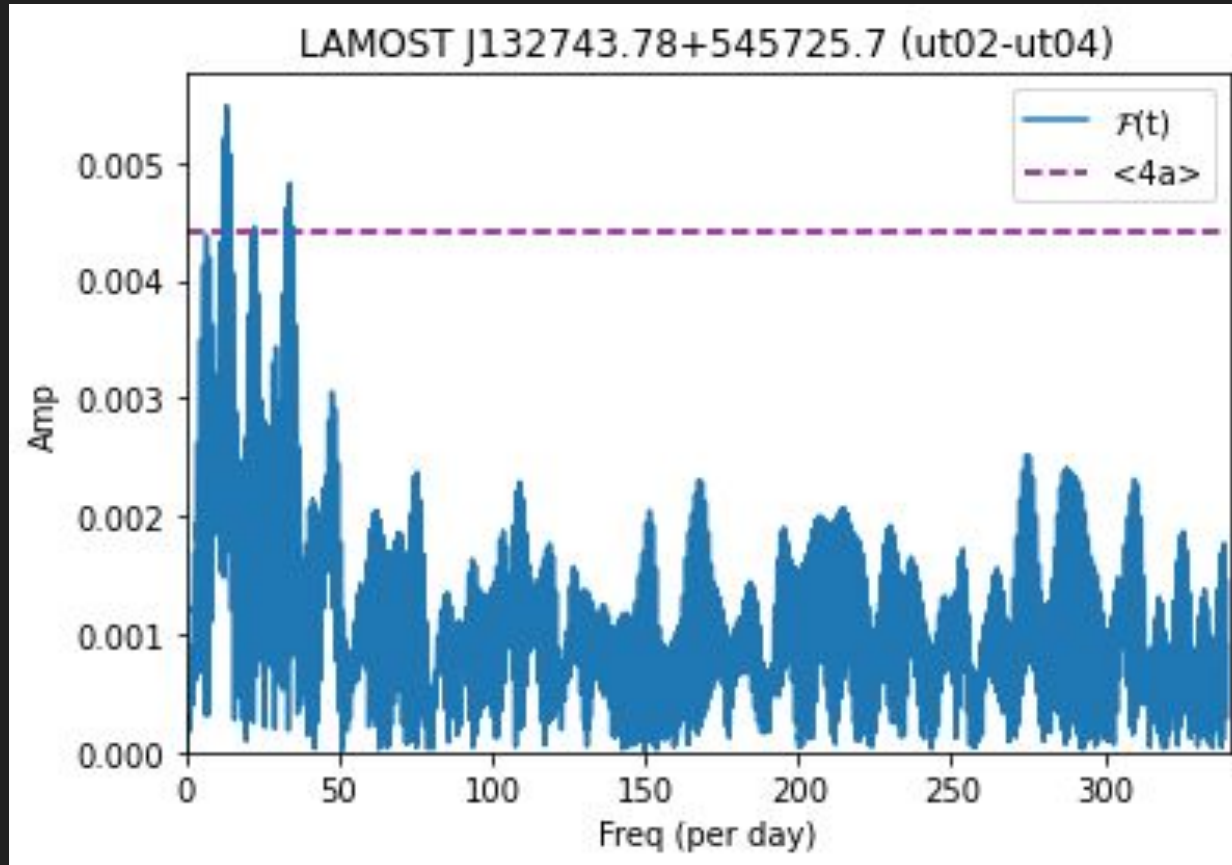


Fourier Transform - UT July 3rd

- Freq: 11.5
- Amp: 0.008

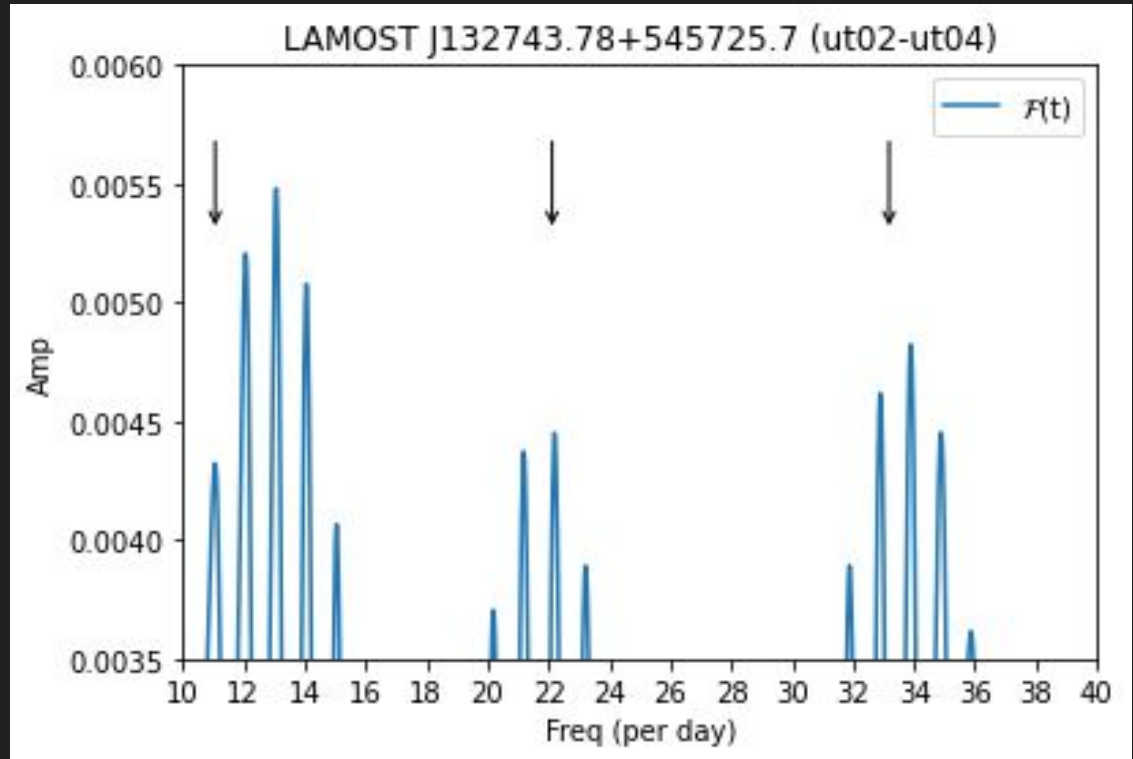


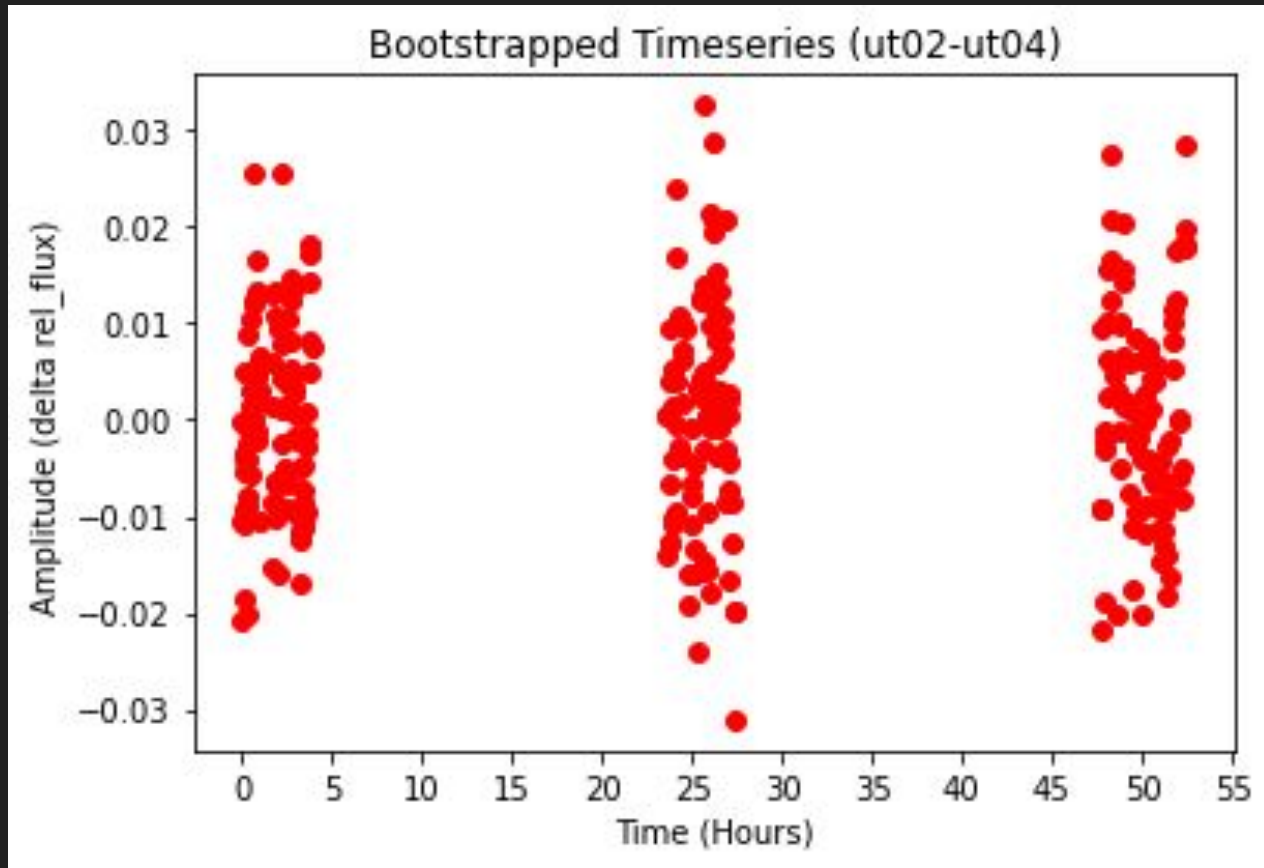
Fourier Transform - UT July 2nd-4th



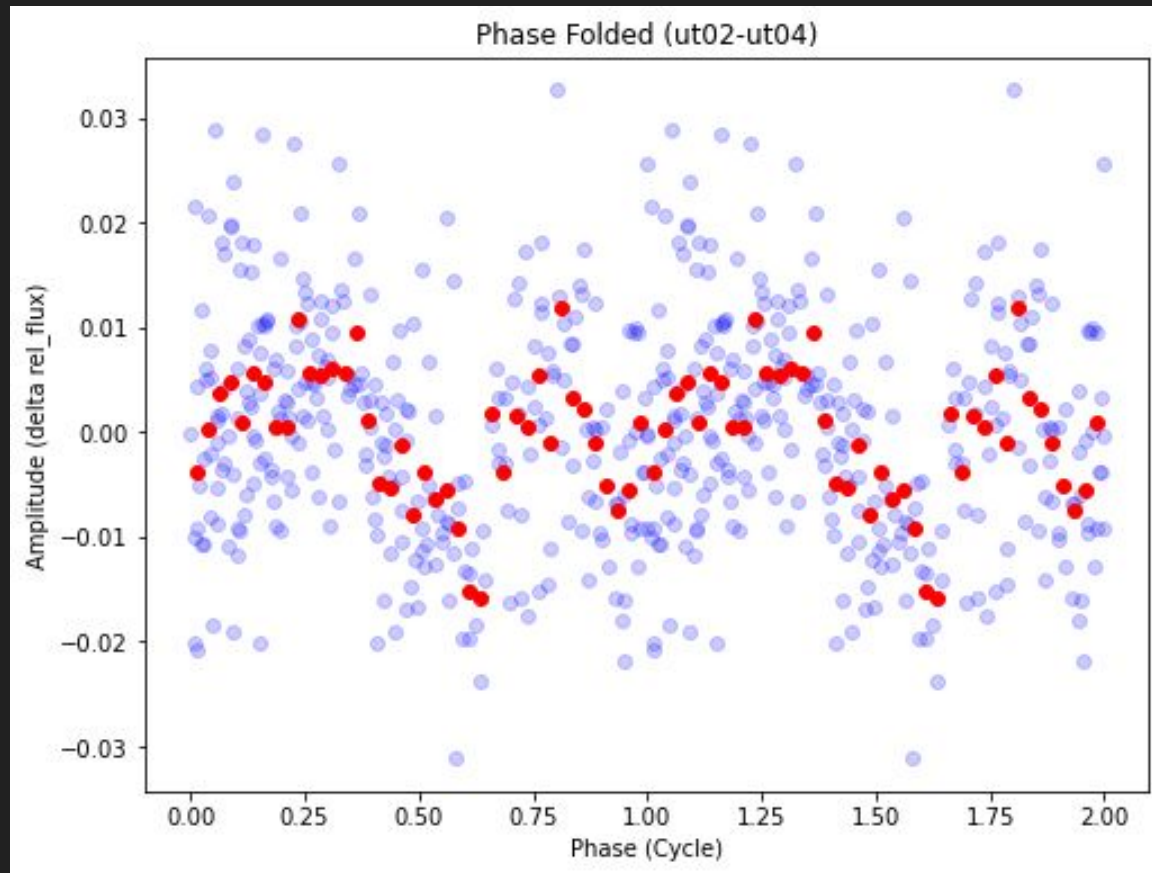
Fourier Transform - UT July 2nd-4th

- Alias 1
 - Freq: 11.06
 - Amp: 0.004
- Alias 2
 - Freq: 12.06
 - Amp: .005
- Harmonics
 - First ≈ 22
 - Second ≈ 33





Source Code used: Dr. Matt Wood



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Results

- PG 1533-057
 - Wasn't able to reduce data UT 06-12-2024
- LAMOST J132743.78+545725.7
 - Frequency: 11.06 ± 0.03 per day
 - Period: 2.170 ± 0.006 hours
- WD 1459+347
 - Not found to vary at the 2% (Amp less than $\langle 4\sigma \rangle$)

Acknowledgments



Dr. Williams



My Peers



Joshua Blue



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